



nexus / **nederland**

PHILIPS

Technical Agreement

Dutch National HIE Network

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Management summary

In the Netherlands there is a growing need for patient related information to be available at the right time at the right place, centered around the individual patient care and crossing the boundaries of single healthcare organizations. Many secondary and tertiary healthcare organizations participate in a “health information exchange” infrastructure. These infrastructures vary as either healthcare organizations are part of an (shared) HIE infrastructure provided by a regional health information exchange organisation, have implemented an exchange infrastructure for themselves, or have adopted a vendor specific exchange solution. Although health information is often successfully exchanged within a health information exchange system there is a commonly felt challenge when the information exchange needs to cross the boundaries of a single health information exchange network or system.

To address these challenges the Dutch national competence center for electronic exchange of health and care information, Nictiz promotes a modified version¹ of the European Interoperability Framework² that identifies six “layers” of interoperability. Each layer representing an area or domain parties that exchange information have to agree on.

This document is the result of a multi-vendor³ initiative under the supervision of the “VNO/NCW Taskforcare Samen Vooruit”. It details the “infrastructure” and “application” layers of the Nictiz Interoperability Model.

This document neither claims to provide all answers to the challenges associated with health information exchange, nor does it deny the fact that there are alternatives to achieve the same goal.

This document does specify the roles and responsibilities a system has that chooses to adopt the IHE Cross-Community Access (XCA) and Cross-Community Access - Images (XCA-I) profiles to exchange health information with another system regardless whether these systems are of the same or different vendors.

¹ <https://www.nictiz.nl/rapporten/electronic-information-for-health-and-care-services/>

² https://ec.europa.eu/isa2/eif_en

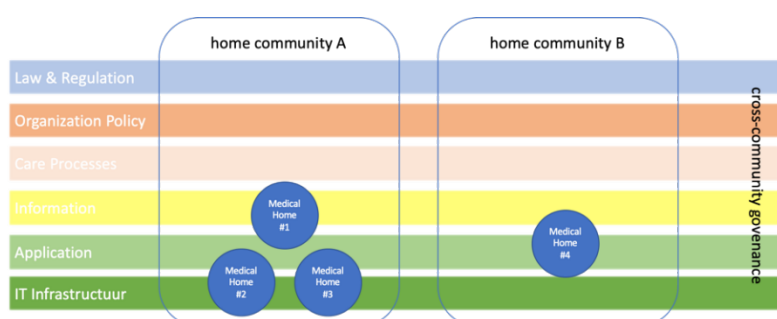
³ Epic, Chipsoft, Nexus-Nederland, Agfa, Enovation, Philips

1. Introduction

This Technical Agreement (TA) describes and specifies technical and quality responsibilities to which parties agree when connecting to the Dutch National HIE Network (DNHN). DNHN stands for the variety of HIE's in the Netherlands that are connected. This document is written in English to allow foreign companies to participate and understand the requirements to allow for integration with the Dutch National HIE Network.

1.1. Background

In an information exchange use-case patient (medical) information can be considered to have a "medical home". This medical home is often represented by a clinical information or HIE system that allows for exchange of the information it is responsible for. One or more "medical homes" can form a "home community". A home community is often managed by an HIE organization (Dutch: RSO) or by a healthcare organization itself. When one or more "home communities" need to share information, they need to do so according to an agreed upon "cross-community governance" (Dutch: afsprakenstelsel) that defines the "rules of engagement".



This document is about the technical agreements that are part of this governance for home communities that participate in the Dutch National HIE Network.

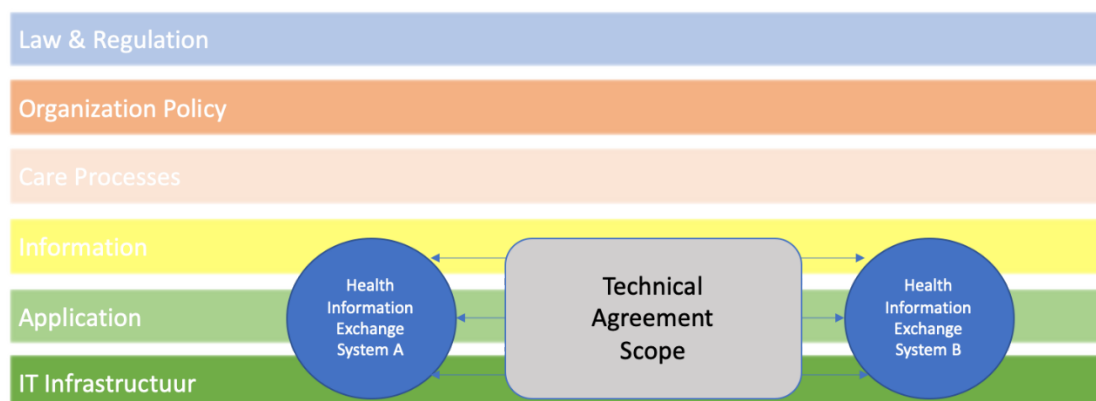
1.2. Definition of terms

Term	Definition
Affinity Domain	An XDS(-i) Affinity Domain is a (group of) healthcare enterprise(s) that has/have agreed to work (together) using a common set of policies and share a common infrastructure.
TA	Technical Agreement.
Health Information Exchange (HIE)	The act of exchanging patient health related information between two or more parties.
HIE System	The system or infrastructure used to share health information.
HIE Vendor	Vendor of an HIE system that is part of an HIE infrastructure and interacts
HIE Administrator	The party responsible for maintaining the HIE infrastructure
home community	A coupling of facilities/enterprises that have agreed to work together using a common set of policies for the purpose of sharing clinical information via an established mechanism.
IHE	Integrating the Healthcare Enterprise
IHE Connectathon	Testing event organized by IHE.
IHE Integration Profile	Implementation guideline how to use a specific communication protocol to share information.
Actor	Role or responsibility a system has in relation to the use-case defined by an IHE Integration Profile
Transaction	Technical definition how information is exchanged between two or more actors
SAML	Security Assertion Mark-up Language
Identity Provider (IdP)	System that establishes electronic identity of a person

1.3. Scope and assumptions

This TA takes the [NICTIZ Interoperability Model](#) as guidance and references the lower two layers (IT Infrastructure and Application) of this model. It touches the information layer but will not go into detail as to “what” medical information will be exchanged.

This TA entitles vendors to make their own choices with respect to how they design or implement their HIE system. This TA only addresses the (boundary) interfaces such systems need to implement when interacting with to other HIE systems in order to exchange (patient) information.



The following principles are used while creating this document:

1. This TA extends existing (decentralized) health information exchange systems and infrastructures.
2. This TA acknowledges that existing systems are build using different technology stacks.
3. This TA does not prevent vendors of HIE systems from competing in the marketplace based on functionality, quality, service and/or price.
4. This TA assumes the exchange of medical documents (e.g. Patient Summaries, (scanned) Letters, (Radiology) reports in CDA document format, medical documents in a binary object format, and medical images (in the DICOM format).
5. This TA assumes that a vendor that represents an HIE is responsible for implementing this TA. If a vendor within the HIE is not compliant with this TA, they can cooperate with another vendor to comply to this TA.

1.4. Relation to other documents

This Technical Agreement covers some aspects discussed in other documents published on the NICTIZ website such as the [“handreiking interoperabiliteit tussen zorginstellingen 2019”](#).

Since the technical agreement will only address the boundary interfaces of an affinity domain, it will only describe the responsibilities of these boundary interfaces. The goal of the technical agreement is not to standardize content but to define what boundary interfaces can expect when integrating in the Dutch National HIE Network and outline the responsibilities of these interface on a more technical level.

1.5. Format of the technical agreement

This Technical Agreement starts by outlining the IHE Integration Profiles and transactions that are applicable within the Dutch National HIE Network. The IHE profiles are the foundation of the network and therefore this technical agreement will refer to them wherever possible.

The document will then outline any technical and non-technical responsibilities of HIE vendors or HIE administrators of an HIE infrastructure. It also addresses the responsibilities the HIE vendor should be aware of for successful integration, but to which the vendor itself can't take responsibility.

Finally, the TA addresses specific technical agreements to which HIE vendors will commit in relation to the transactions of the boundary interfaces (XCA(-i) gateways).

2. IHE standards and transactions

This TA applies on the boundary interfaces of home communities. Only transaction between the boundary actors (XCA(-i) gateways) will be addressed. In all cases the standards as described by IHE are leading. There were these leave room for interpretation this TA will address the agreements vendors will commit to. These explicit agreements described in the chapter Technical Agreements.

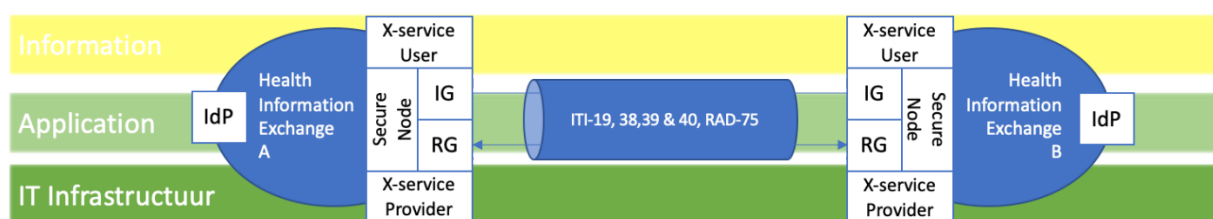
2.1. IHE mandatory transactions

The TA mandates the use of the IHE XCA, XCA-I and XUA profiles for the interfaces through which the HIE systems interact. Hence, HIE system vendors have the obligation to implement the IHE Initiating Gateway, Responding Gateway, X-Service User and X-Service Provider actors. For security purposes the IHE XUA and ATNA profiles and their actors are used.

In order to fulfil the (XUA) X-Service User actor an HIE system must be able to provide a trusted user identity. How this IdP responsibility is achieved by the HIE system is out of scope for this TA. This TA will only dictate the minimal set of “user attributes” (or claims) the identity provider must be able provide. The IHE profiles this TA references can be found in:

- IT Infrastructure Technical Framework volume 1 t/m 3.
- Radiology Technical Framework volume 1 & 2.

Unless otherwise specified the latest version of the above listed frameworks are referenced.



An HIE Vendor shall implement, and successfully validate at an IHE Connectathon, the following actors & transactions within their HIE system:

Actor	Transaction	Profile
Secure Application Secure Node	ITI-19 (Node Authentication)	ATNA
Initiating Gateway Responding Gateway	ITI-38 (Cross-community Query) ITI-39 (Cross-community Retrieve)	XCA
Initiating Imaging Gateway Responding Imaging Gateway	RAD-75 (cross-community Retrieve Imaging Document Set)	XCA-I
X-Service User X-Service Provider	ITI-40 (Provide X-User Assertion)	XUA

2.2. Other (optional) transactions

2.2.1. WADO WS

Since the above transactions only allow for the exchange of documents and full DICOM studies the optional transaction “WADO-WS” has been implemented. WADO-WS is not an official IHE profile, but due to the need of exchanging lossy compressed images via XCA-I the vendors VANAD Enovation and Philips Interoperability solutions (Forcare at the time) decided to support this standard during the Dutch IHE steering group consultation of November 27th.

Referenced documentation for the WADO-WS transaction can be found on:

- https://wiki.ihe.net/index.php/WADO-WS_Extensions_for_XDS-I.b_and_XCA-I
- http://dicom.nema.org/dicom/2013/output/chtml/part18/sect_6.4.html

Actor	Transaction	Profile
Initiating Imaging Gateway	WADO-WS	DICOM
Responding Imaging Gateway	Retrieve Rendered Imaging Document Set	WADO-WS

Implementation of the WADO-WS transaction shall be agreed upon on a case by case bases since this transaction strongly relies on the capabilities of both the Imaging Source and the XDS(-i) consumer. Since the WADO-WS transaction is not tested on an IHE Connectathon it shall therefor be tested on an acceptance environment with at least two other vendors that are part of the Dutch National HIE Network. To allow for integration of this transaction by HIE Vendors the transactions specifications are part of this TA agreement. The (reference to the) WSDL can be found in the appendix.

3. Technical responsibilities

3.1. Network security

Connecting HIE vendors must assure the following regarding network security.

3.1.1. Connectivity on infrastructure level

The administrator of an HIE infrastructure is responsible for allowing other parties to connect to the endpoints as provided in the "Connection Document". If there any special requirements for achieving connectivity (e.g. IP whitelisting, IP-sec tunnelling etc.) the administrator of an HIE infrastructure shall mention these requirements in the connection document.

3.1.2. Domain Name System addressing endpoints

The administrator of an HIE infrastructure shall provide a Domain Name System (DNS) Address endpoint of the XCA(-i) actor and maintain the DNS entries that allow connecting to these endpoints.

3.1.3. Bandwidth

The administrator of an HIE infrastructure is responsible for sufficient bandwidth from and to the HIE system in order to support the exchange of medical information. In case of XCA-I transactions (which allows for the exchange of DICOM studies) the HIE system is preferred to have a 1 Gigabit network connection (1,000 megabits) to the HIE system(s) involved with these transactions.

3.1.3.1. Server certificates

- In conformance with the [ATNA](#) profile all XCA(-i) gateways must implement the secure node actor providing secure network connections through mutual TLS. Endpoints of the secure nodes shall therefore always use the Hypertext Transfer Protocol Secure (https) protocol.
- On production environments the server certificate is provided specifically for that server.
- Server certificates used for mutual TLS shall be signed by a publicly trusted certificate authority.
- HIE infrastructures can use a loadbalancer set-up. How load balancing is implemented or set up technically is outside the scope of this document and remains the HIE vendors choice. The loadbalancer and/or the load balancing setup however shall support establishing mutual TLS. All certificates required to establish mutual TLS, for the transactions specified in this document, will be part of the connection document and will comply to the points above.

3.2. Performance

By default, the Dutch National HIE Network relies on a synchronous message exchange. This requires that the processing of the request and formulation of the response needs to take place within the boundaries of the HTTP timeout value. Therefor vendors of HIE systems have a best-efforts obligation towards maintaining the performance of the IHE infrastructure.

Vendors agree that typically processing a request and start formulating an initial response should take no longer than 10 minutes, which can then be considered as the “default timeout boundary” especially considering the exchange of DICOM images (specifically the WADO-WS and RAD-75 transactions). If the usecase allows for a lower HTTP time-out boundary, this can be applied by vendors. For example, an usecase where only documents are exchanged via ITI-38 and ITI-39. Note that the HTTP timeout should be in line with the Mutual TLS timeout setting (5.2.1.1 (Mutual) TLS timeout settings).

If an affinity domain repeatedly fails to respond within this 10 minute “default timeframe”, an HIE vendor must be able to allow an HIE administrator to configure a higher time-out value. In case time-out exceeds the agreed upon threshold the initiating gateway can close the connection.

HIE vendors will investigate the performance of the affinity domain and to their best efforts identify any HIE system that might be the cause of the delay. If the performance is impacted by systems controlled by the HIE vendor, the HIE vendor shall, to their best effort try to mitigate the performance issues in order to maintain the timeout window that is acceptable for involved parties.

How this best effort translates directly to potentials solutions, such as load balancing or allocating additional virtual or physical system resources to an HIE system, is out of scope for this TA allowing vendors to make their own choices.

It is possible that the performance is impacted by systems beyond the boundary interfaces that are not controlled by the HIE vendor. In that case the HIE vendor shall inform the involved parties about the set timeout boundaries and inquire if there is a way that these boundaries can be met. If not, parties can agree on setting new boundaries that are acceptable for all involved parties.

3.3. Availability of services

Availability agreements are out of scope for this technical agreement. These agreements are customer facing only and do not apply on the boundary interfaces discussed in this TA.

However, HIE vendors shall to their best-effort maintain the highest possible availability for both acceptance as production environments allowing for the best possible 24/7 nationwide availability of medical information for affinity domains connected to the Dutch National HIE Network.

3.4. Auditing

HIE vendors shall provide a means for Accountability, meaning each transaction as mentioned in chapter “IHE standards and transactions” shall be logged, allowing for a full audit of the integration. This can be achieved in several ways and is not limited to implementation of the IHE ATNA profile.

Systems that are responsible for processing auditing messages from the HIE systems can expect that audit messages contain the information as was exchanged between these boundary interfaces. This in conformance with the ATNA specifications as defined in chapter “3.40.4.2 ATNA Audit encoding”.

When an initiating gateway is requested to resign a SAML token the responding gateway shall use information from the resigned token for its ATNA audit logging.

3.5. Consistent time

HIE systems shall implement the [consistent time](#) profile.

3.6. Respecting cooperation agreements

An initiating gateway shall respect cooperation agreements and only query home communities that contain information for which a cooperation agreement is available.

An XCA(-i) responding gateway can act as an entry point to multiple home communities / organizations. It is the responsibility of the queried home community to respect the “cooperation agreements”.

How this is achieved and implemented is the choice of an HIE vendor. The information in the XUA (SAML) token as described in chapter 8.2 *XUA related agreements* contains the required information to enforce these cooperation agreements.

3.7. Applying patient consent

It's a medical record custodians' responsibility to enforce the patient's consent. This technical agreement does not dictate how or where within the affinity domain an HIE vendor should implement rules that apply patient consent to incoming queries.

4. Non-technical responsibilities

4.1. Responsibilities vested within the HIE providing organization

The following responsibilities can impact the transactions between affinity domains. These responsibilities are not necessarily that of the HIE vendor but need to be aligned on clearly within the affinity domain that the HIE vendor represents. HIE vendors integrating within the Dutch National HIE Network can assume that these responsibilities are taken care of within the affinity domain it is connecting to.

4.1.1. Cooperation agreements

The topics below are typically part of a cooperation agreement (Dutch: samenwerkingsovereenkomst). This agreement is relevant when integrating with the Dutch National HIE Network as these can impact the integration between the home communities.

4.1.1.1. User and user roles

Within the scope of the technical agreement it is relevant that the cooperation agreement states:

- That users are individually traceable and authenticated based on a personal account
- Which user roles are allocated on user authentications and which user roles can query and retrieve information from other home communities?

4.1.1.2. Patient identification and BSN verification

Within the scope of the technical agreement it is relevant that both the initiating gateway and the responding gateway can assume that it is the medical record custodians' responsibility that the patient was identified, and the BSN was verified by legal standards. Where applicable this verified BSN is used as the patient ID in the transactions described in this document.

4.1.1.3. Patient consent registration and consent scope

Within the scope of the technical agreement it is relevant that the cooperation agreement states that it is the responsibility of the medical record custodian to capture a patient's consent and only share medical information in accordance with the patient's consent. How this responsibility is (technically) fulfilled is up to the medical record custodian and the HIE administrator, within the possibilities offered by the HIE vendor.

4.2. Legal compliance

This technical agreement does not address legal compliance that may be applicable to the context in which patient information is exchanged between two (or more) medical record custodians. It is assumed that all parties involved in the exchange obey to the applicable legislation(s).

4.3. IHE integration statements

Organizations and/or vendors providing a gateway service to other HIEs shall be able to exchange an IHE conformance statement of their gateway service.

4.4. IHE Connectathon participation and scope

Organizations and/or vendors providing a gateway service shall use a system that successfully validated the transactions mentioned in chapter 2.1 *Mandatory IHE transactions* at an IHE. Such Connectathon result shall be publicly available at the IHE Connectathon Results Website.

4.5. Information quality and metadata

The responsibility of the information quality and the meta-data lies outside the scope of these boundary interfaces.

4.6. Providing a connection document

Organizations and/or vendors providing a gateway service shall be responsible to provide a "connection document" through which the technical (connectivity) details for other HIEs are made available. A template connection document is added as an appendix^[6].

5. Technical agreements

5.1. Generic IHE profile related agreements

5.1.1.1. Synchronous vs Asynchronous messaging

As [described](#) by IHE, when two actors, referred to as Requestor and Provider, need to exchange web services based messages using a request-response message exchange pattern, they can do so synchronously or asynchronously. This technical agreement allows for vendors to integrate using either a synchronously as well as asynchronously exchange of messages in conformance with IHE specifications and the Asynchronous Web Services Exchange Supplement.

However, in practice, the Dutch National HIE Network relies on synchronous message exchange since it is expected that the processing of the request and formulation of the response takes very little time and lies well within the boundaries of the HTTP timeout values as described in the paragraph "Infrastructure and security agreements". Therefore, an HIE vendor shall support synchronous messaging.

5.2. Infrastructure and security agreements

5.2.1. Secure Node authentication

5.2.1.1. (Mutual) TLS timeout settings

For node authentication TLS inactivity timeout settings should be set as low as possible but in accordance with the HTTP timeout values set in the HIE infrastructure. Note that in case of a DICOM exchange usecase a minimum HTTP timeout value of 10 minutes is defined as "normal" (chapter 4.2). The TLS inactivity timeout settings shall be aligned with the XCA-I HTTP time out values when (large) DICOM studies are retrieved.

5.2.1.2. Mutual TLS requirements

- The HIE vendor shall use a TLS version qualified as good (in Dutch: goed) or sufficient (in Dutch: voldoende) in conformance with the guidelines ([ICT-beveiligingsrichtlijnen voor Transport Layer Security \(TLS\)](#)) of the [National Cyber Security Center](#) (NCSC).
- The HIE vendor shall apply hashing standards qualified as good (in Dutch: goed) or sufficient (in Dutch: voldoende) in conformance with the above-mentioned guidelines
- The HIE vendor shall use cypher suites qualified as good (in Dutch: goed) or enough (in Dutch: voldoende) in conformance with the above-mentioned guidelines

5.2.1.3. Certificate requirements

- The certificate authority of a server certificate should be a known trusted root (or intermediate) authority for the clients
- The certificate shall not be a self-signed certificate
- The certificate used for authentication can be used for both server side and client-side authentication.
- The SubjectDN/CN of a system certificate shall match the system's public host name.
- The HIE vendor shall provide the full certificate chain including all CA's between the root CA and the certificate

5.3. ITI-40 XUA related agreements

The XUA profile carries a readable and verifiable claim of the user identity, authentication method, and as needed their roles, purpose of use, and consent. This chapter outlines any specific agreements regarding this profile and dealing with multiple identity providers.

5.3.1.1. Dealing with multiple Identity providers

Within a home community there can be multiple ways a user is authenticated, therefore there can be a variety of identity providers that are responsible for the claim of the user's identity. The technical agreement identifies two ways of dealing with this situation.

5.3.1.2. Resigning of the XUA token (preferred)

HIE vendors of the systems that act as the initiating gateway actor may modify a SAML token and shall resign this token if modified.

Resigning a XUA token may be done to achieve operational efficiencies and network scalability. In case this is done, the Initiating Gateway actor is responsible for resigning the XUA token that was created by any of the trusted identity providers within the home community. The consequence of resigning the XUA token is that the Responding Gateways trusts all identities of the initiating community based on the resigning authority.

In this case:

- HIE vendors of the systems that acts as the initiating gateway actor shall provide **one** identity provider certificate and URI and provide the required details in their connection document
- HIE vendors of the system that acts as the responding gateway actor shall configure the responding gateway to trust this identity provider

5.3.1.3. Forwarding the claim of the original identity provider

An HIE vendors can choose to forward a claim from the initiating gateway without modifications or resigning of the claim. This however creates the need for the responding gateway to trust all identity providers that reside within the home communities the initiating gateway represents. In this case it is agreed that:

- HIE vendors of the system that acts as the initiating gateway actor shall provide **all** identity provider certificates and URI details in their connection document
- HIE vendors of the initiating gateway shall not change any of the claims made by the original identity provider
- HIE vendors of the system that acts as the responding gateway actor shall configure the responding gateway to trust all provided identity providers

5.3.1.4. SAML Token attributes (claims)

The following table is derived from *ITI TF Vol. 2b section 3.40.4.1.2 (Message Semantics)* and lists the SAML token attributes allowed to be included.

Attribute	R/O/C	Name space	Description
Subject ID	O	urn:oasis:names:tc:xspa:1.0:subject:subject-id	The value on the Subject ID attribute shall be a plain text description of the user's name (not user ID) . This SubjectID may be used for audit logging by the responding gateway.
Subject Organization Name	C	urn:oasis:names:tc:xspa:1.0:subject:organization	The value on Subject Organization attribute shall be a plain text description of the organization. This Subject Organization Name may be used for audit logging by the responding gateway. If no Subject Organization name is available, a Subject Organization ID shall be provided.
Subject Organization ID	C	urn:oasis:names:tc:xspa:1.0:subject:organization-id	A unique identifier for the organization that the user is representing in performing this transaction. If no Subject Organization ID is available, a Subject Organization Name shall be provided.
Home Community ID	O	urn:ihe:iti:xca:2010:homeCommunityId	The value shall be the Home Community ID (an Object Identifier) assigned to the Community that is initiating the request.
National Provider Identifier	O	urn:oasis:names:tc:xspa:1.0:subject:npi	A National Provider Identifier (NPI) is a unique identifier issued to health care providers by their national authority.
Other Provider Identifier Attribute	O	urn:ihe:iti:xua:2017:subject:provider-identifier	A unique identifier issued to health care providers by a named authority. This attribute may be repeated.
Subject-Role	R	urn:oasis:names:tc:xacml:2.0:subject:role	The role attribute shall contain one or more role codes from the identified Value-Set that represents the role that the XUA user is playing when making the request.
Authz-Consent	O	urn:ihe:iti:bppc:2007:docid or urn:ihe:iti:xua:2012:acp	Conveys either the BPPC document relevant for this transaction context or it references the related Privacy Policy ID agreed upon between communities.
Patient Identifier	O	urn:oasis:names:tc:xacml:2.0:resource:resource-id	Shall be the patient's BSN for which the transaction is executed.
Purpose of Use	O	urn:oasis:names:tc:xspa:1.0:subject:purposeofuse	The PurposeOfUse element shall contain the coded representation of the Purpose for Use that is in effect for the request. The X-Service Provider may use the PurposeOfUse value in Access Control decisions. In case no PurposeOfUse element is present in the transactions; the custodian of the medical record can apply its local policies.

5.3.1.5. Subject Organization Name attribute

The Subject Organization Name attribute shall be Descriptive Name as listed in the HL7 Nederland OID register (<https://www.hl7.nl/component/phocadownload/category/13-oid.html>).

Otherwise they shall use a representative Descriptive Name, such as the name that is found when looking up the Subject Organization ID attribute in its corresponding register.

5.3.1.6. Subject Organization ID attribute

The Subject Organization ID attribute shall be the OID as listed in the HL7 Nederland OID register (<https://www.hl7.nl/component/phocadownload/category/13-oid.html>).

Otherwise a Subject Organization ID shall be used that is traceable and verifiable in a publicly available register such as the AGB register.

5.3.1.7. Home Community ID attribute

The Home Community ID attribute shall be the OID of the home community as provided in the connection statement.

5.3.1.8. Subject-Role attribute

For a user one or more roles can be provided in the subject role attribute.

As described in Vol. 2b - Section 3.40.4.2 - Additional section to add to ALL ATNA audit messages when the transaction includes XUA Assertion. The iDP is responsible for Subject-Role claim.

The HIE system shall be able to interpret or translate the roles coming from the Initiating Gateway. If no national standards are defined or implemented regarding the Subject-Role claim, a variation of roles can be expected.

To support the use case of exchanging medical information between one or more home communities, we define that the initiating HIE system shall be able to translate the Subject-Role claim to at least one of the roles as described in the table below.

Subject-Role	Description
MedicalDoctors	A doctor
Assistants	Any kind of assistant of to a medical doctor
PrivacyOfficers	A privacy and security officer
Administrators	A system administrator

If a use case requires additional Subject-Role claims the use and interpretation of these claims lies outside of the scope of this technical agreement.

5.3.1.9. Patient Identifier Attribute

In cases where a BSN is provided it shall be provided as a HL7 CX Data Type (as defined in Vol. 2b - Section 3.40.4.1.2.2.1 Patient Identifier Attribute) with at a minimum the follow components specified:

- ID Number
- Assigning Authority

Example:

123456789^^^&2.16.840.1.113883.2.4.6.3&ISO

5.4. ITI-38 transaction agreements

The ITI-38 transaction shall contain an IHE-XUA token.

5.5. ITI-39 transactions agreements

The ITI-39 transaction shall contain an IHE-XUA token.

5.6. RAD-75 transactions agreements

The RAD-75 transaction shall contain an IHE-XUA token.

5.7. WADO-WS transactions agreements

The WADO-WS transaction shall contain an IHE-XUA token.

6. Involved parties

This document is a co-creation of Enovation and Philips Interoperability Solutions.
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7. Version control

Rev	Release Date	Author	Description of changes
0.10	17-SEP-2020	Rob Treffers	Draft, initial version
0.50	22-SEP-2020	Rob Treffers	Processed input Andries Hamster
0.60	01-OCT-2020	Rob Treffers	Discussion with involved parties.
0.70	02-OCT-2020	Rob Treffers	Discussed document up till chapter 5.2.
0.75	08-OCT-2020	Alex Settels	Discussed document up till end
0.76	13-OCT-2020	Rob Treffers	Processed remarks and final comments
0.77	02-NOV-2020	Rob Treffers	Included external remarks as comments.
0.78	03-NOV-2020	Rob Treffers	Processed remarks from Agfa, Epic and Nexus Nederland up to end of document. Left comments open and make reference to excel response to make changes clear.
0.80	03-NOV-2020	Rob Treffers	Accepted the changes corresponding the comments based on remarks from Agfa, Epic and Nexus. Removed comments. Included connection document template
0.81	05-NOV-2020	Rob Treffers	Included remarks from Chipsoft in comments.
0.82	06-NOV-2020	Rob Treffers	Processed remarks from Chipsoft up to end of document. Left comments open and make reference to excel response to make changes clear.
0.83	06-NOV-2020	Rob Treffers	Accepted changes and removed comments. For review version
0.85	17-NOV-2020	Andries Hamster	Included management summary
0.86	19-NOV-2020	Rob Treffers	Included remarks from Agfa, Epic and Nexus Minor change to management summary
0.87	23-NOV-2020	Rob Treffers	Included informal feedback from VWS
1.00	23-NOV-2020	Rob Treffers	Processed remarks, removed comments, accepted changes. Additional layout changes. Changed classification and document status. Included WADO-WS WSDL
1.10	16-FEB-2020	Rob Treffers	Edited first sentence of management summary. 5.3.1.4 SAML Token attributes: Fixed typo in homecommunity id attribute.

A.1.1 Environments

A.1.2 Endpoints

A.2.1 Acceptance

(**<PEM certificate>**) example:

-----BEGIN CERTIFICATE-----

MlFXTCCEBWEgAwBlAgIQCpqbgZuqS3J828pW3poelSTANBgkqhkiG9w0BAQsFADBGMQswCQYDVQQGEwJVUzEPMA0GA1UEChMGQW1hem9uMRUwEwYDVBQLExwTZXJ2ZXIgQ0EGMUlxZDZANBgNVBAMTBKFTyXpvbjAeFw0yMDA5MDQwMDAwMDBaFw0yMT6EMWYDMDAwMDBABASXBjMBAGNBVAMTBW51Lm5sMlBIlANBgkqhkiG9w0BAQEFAAOCAQMIBCGkCAQEAp4zB4CLBXDKiU0A5CWpdPnS755L2j95XCdv396vaoxcEgx3JU/plysP5x6asHeJPqPbb9ITX9OST527oLIY9KEJA9VVJrrr2lnEpGH4CDpMEXIMhWK6pVSK7+S7ev89nwzArJEBeVFUUKmkXdi1CdhdZl9zv7kBFB6T0m3XYiberzrE8XqbEAOKY7eoqe7HYTHvb41USywThzvHo+sbEOJRvu/N2mDy3Km8VE7w9ACMeUQDqKRb98/g33/Uc+VNCEfIR2SwU8LrxCoz/3pdfxHXRlsapvQLdJj+k6BKbpmlv9zl9/3+cdwZDi8wLK3Ys/FT1pAA0k0oThXswIDAQABo4ICezCCAncwHwYDVR0JBggwFoAUUwRmBlKg5WSPKOBYueWAdFv5PdAwHQYDVROBBYEFCQBBSJ2ZF8imCPj+tGJUCYQAQoeMBKGA1UdeEQSMBCCBW51Lm5sggcqLm51Lm5mA4GA1UdDwEB/wQEAwIFDAQwNHSUeFajUBggrBgEFBQCqDAQYIKwYBBQUHAwIwCwYDVR0BDQwMjAwOjC6gLIYqaHR0cDovL2Nybc5zY2ExYi5hbWF6b250cnVzdC5jb20vc2NhMWluY3JsMCAGA1UdIAQZMBcwCwYJYIZIAybnbAECMAGGBmeBDAECA TB1BggrBgEFBQCcBAQRpMGcwLQYIKwYBBQUHMAAGGIWh0dHA6Ly9vY3NwLnNjYTFlmFtFYXpvbnRydXN0LmNvbTA2BggrBgEFBQCwAoYqahr0cDovL2Nydc5zY2ExYi5hbWF6b250cnVzdC5jb20vc2NhMWluY3J0MAwGA1UdEwEB/wQCMAAwwgEFBgorBgEEADZSAglAQCBIIHB2IHZApeAdwD2XJqv0XcwlhRUAGalWfAo400TGTO/3wwwIvAmTvfk4wAAAXRW7gUGAAAEawBIMEYCIQDYQJaAyTJ7K8ZI U23p3kmmlK4aav4ihD3e7f+2T016FGlhAPDX7XiItbjqHzlrWCWxoWcsrKloX6+WxFW03SAZCBEAHYAAXNdKv7mq0VEsV6a1FbmEDf71fpH3KFzLLJe5vbHDsoAAAAF0vU4e/wAABAMARzBFAlAJUWj3BgBBymZZEw0SFxDUTlbm+/GGMJgbfZGfotJzNQhEAKfBZAReigj4lrSi3i9yPwRxWqODOLyVLzTpIlW1e1UMA0GCSqGSlib3DQEBcUA4A4BAQcQdY06wnmlD4lyIrxuSGITOu7Q6FLFRoV7liWychwmzg6K+CbnO05kdEBERmB95tZgvR8HLsTNMAHqPlzh06vOzcwwMjA rNkj p4xoUEgXYAv8/uzMDQpqe17J2iU7frcbM+byXpMQERb7tzps2Nq34OM7Ov954X66gnQZ+3rRuXuuThoin5FypLa8ulFmOm +Ur/dBiKM9qDMTqmMTEqPOMP4m0wbDI3s6819yzggG/PUUm dQxsmpIm7YZFo6rp05Uri9vU+BIKz dqPkhumzamN+e1XcBz8 1ct0ZnrKEOYMdeAGUbZq/pTa8/2HBs0232vEc6vPpnNSAi5ft5uNY25ygQ

-----END CERTIFICATE-----

A.2.2 Production

(**<Issuer>**) and (**<PEM certificate>**)

A.3.1 Acceptance

(<Issuer>) and (<PEM certificate>)

A.3.2 Production

(**<Issuer>**) and (**<PEM certificate>**)

A.4 XCA(-i) Integration details

A.4.1 Acceptance

OID Value	Description
	HomeCommunityID
	Repository Unique ID

Imaging source details (*repeat for each imaging source*)

Value	Description
	Imaging Document Source ID*
	AE Title

**can also be called the "Image retrieve location UID".*

A.4.2 Production

OID Value	Description
	HomeCommunityID
	Repository Unique ID

Imaging source details (*repeat for each imaging source*)

Value	Description
	Imaging Document Source ID*
	AE Title

**can also be called the "Image retrieve location UID".*

A.5 IHE XUA Attributes

The following (additional) SAML attributes are used in the transactions.

SAML2 attribute	Value	Example

B. Appendix: WADO WS specification (WSDL)

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- This wsdl file is for an XDS-I.b Imaging Document Source Actor
It can be used 'as is' to support Retrieve Imaging Document Set Transaction [RAD-69]
using Synchronous Web Services.-->
<definitions name="ImagingDocumentSource"
targetNamespace="urn:ihe:rad:xdsi-b:2009"
xmlns="http://schemas.xmlsoap.org/wsdl/"
xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
xmlns:soap12="http://schemas.xmlsoap.org/wsdl/soap12/"
xmlns:wsaw="http://www.w3.org/2006/05/addressing/wsdl"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:tns="urn:ihe:rad:xdsi-b:2009"
xmlns:wadows="urn:dicom:wado:ws:2011"
xmlns:deprecatedwadows="urn:dicom:ws:wado:2011"
xmlns:ihe="urn:ihe:iti:xds-b:2007"
xmlns:rs="urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0"
xmlns:lcm="urn:oasis:names:tc:ebxml-regrep:xsd:rim:3.0">
<documentation>IHE XDS-I.b Imaging Document Source</documentation>
<types>
<xsd:schema elementFormDefault="qualified">
<xsd:import namespace="urn:oasis:names:tc:ebxml-regrep:xsd:rs:3.0" />
<xsd:import namespace="urn:ihe:iti:xds-b:2007" />
<xsd:import namespace="urn:ihe:rad:xdsi-b:2009" />
</xsd:schema>
</types>
<message name="RetrieveImagingDocumentSetRequest_Message">
<documentation>Retrieve Imaging Document Set</documentation>
<part name="body" element="tns:RetrieveImagingDocumentSetRequest" />
</message>
<message name="RetrieveRenderedImagingDocumentSetRequest_Message">
<documentation>Retrieve Rendered Imaging Document Set</documentation>
<part name="body" element="wadows:RetrieveRenderedImagingDocumentSetRequest" />
</message>
<message name="DeprecatedRetrieveRenderedImagingDocumentSetRequest_Message">
<documentation>Deprecated Retrieve Rendered Imaging Document Set</documentation>
<part name="body" element="deprecatedwadows:RetrieveRenderedImagingDocumentSetRequest" />
</message>
<message name="RetrieveRenderedImagingDocumentSetResponse_Message">
<documentation>Retrieve Rendered Imaging Document Set Response</documentation>
<part name="body" element="wadows:RetrieveRenderedImagingDocumentSetResponse" />
</message>
<message name="RetrieveDocumentSetResponse_Message">
<documentation>Retrieve Document Set Response</documentation>
<part name="body" element="ihe:RetrieveDocumentSetResponse" />
</message>
<portType name="ImagingDocumentSource_PortType">
<operation name="ImagingDocumentSource_RetrieveImagingDocumentSet">
<input message="tns:RetrieveImagingDocumentSetRequest_Message"
wsaw:Action="urn:ihe:rad:2009:RetrieveImagingDocumentSet" />
<output message="tns:RetrieveDocumentSetResponse_Message"
wsaw:Action="urn:ihe:iti:2007:RetrieveDocumentSetResponse" />
</operation>
<operation name="ImagingDocumentSource_RetrieveRenderedImagingDocumentSet">
<input message="tns:RetrieveRenderedImagingDocumentSetRequest_Message"
wsaw:Action="urn:dicom:wado:ws:2011:RetrieveRenderedImagingDocumentSet" />
<output message="tns:RetrieveRenderedImagingDocumentSetResponse_Message"
wsaw:Action="urn:dicom:wado:ws:2011:RetrieveRenderedImagingDocumentSetResponse" />
</operation>
<operation name="ImagingDocumentSource_DeprecatedRetrieveRenderedImagingDocumentSet">
<input message="tns:DeprecatedRetrieveRenderedImagingDocumentSetRequest_Message"
wsaw:Action="urn:dicom:ws:wado:2011:RetrieveRenderedImagingDocumentSet" />
<output message="tns:RetrieveDocumentSetResponse_Message"
wsaw:Action="urn:ihe:iti:2007:RetrieveDocumentSetResponse" />
</operation>
</portType>
<binding name="ImagingDocumentSource_Binding" type="tns:ImagingDocumentSource_PortType">
<soap12:binding style="document" transport="http://schemas.xmlsoap.org/soap/http" />
<wsaw:UsingAddressing wsdl:required="true" />
<operation name="ImagingDocumentSource_RetrieveImagingDocumentSet">
<soap12:operation soapActionRequired="false" />
<input>
<soap12:body use="literal" />

```

```
</input>
<output>
  <soap12:body use="literal" />
</output>
</operation>
<operation name="ImagingDocumentSource_RetrieveRenderedImagingDocumentSet">
  <soap12:operation soapActionRequired="false" />
  <input>
    <soap12:body use="literal" />
  </input>
  <output>
    <soap12:body use="literal" />
  </output>
</operation>
<operation name="ImagingDocumentSource_DeprecatedRetrieveRenderedImagingDocumentSet">
  <soap12:operation soapActionRequired="false" />
  <input>
    <soap12:body use="literal" />
  </input>
  <output>
    <soap12:body use="literal" />
  </output>
</operation>
</binding>
<service name="ImagingDocumentSource_Service">
  <port name="ImagingDocumentSource_Port_Soap12" binding="tns:ImagingDocumentSource_Binding">
    <soap12:address location="http://servicelocation/ImagingDocumentSource_Service" />
  </port>
</service>
</definitions>
```

End of Document